intervening gas clouds. Every known quasar, no matter where on the sky it's found, shows features from dozens of isolated hydrogen clouds scattered across time and space. This unique class of intergalactic object was first identified in the 1980s, and continues to be an active area of astrophysical research. Where did they come from? How much mass do they all contain?

Every known quasar reveals these hydrogen features, so we conclude that the hydrogen clouds are everywhere in the universe. And, as expected, the farther the quasar, the more clouds are present in the spectrum. Some of the hydrogen clouds (less than one percent) are simply the consequence of our line of sight passing through the gas contained in an ordinary spiral or irregular galaxy. You would, of course, expect at least some quasars to fall behind the light of ordinary galaxies that are too distant to detect. But the rest of the absorbers are unmistakable as a class of cosmic object.

Meanwhile, quasar light commonly passes through regions of space that contain monstrous sources of gravity, which wreak havoc on the quasar's image. These are often hard to detect because they may be composed of ordinary matter that is simply too dim and distant, or they may be zones of dark matter, such as what occupies the centers and surrounding regions of galaxy clusters. In either case, where there is mass there is gravity. And where there is gravity there is curved space, according to Einstein's general theory of relativity. And where space is curved it can mimic the curvature of an ordinary glass lens and alter the pathways of light that pass through. Indeed, distant quasars and whole galaxies have been "lensed" by objects that happen to fall along the line of sight to Earth's telescopes. Depending on the mass of the lens itself and the geometry of the line-of-sight alignments, the lensing action can magnify, distort, or even split the background source of light into multiple images, just like fun-house mirrors at arcades.

One of the most distant (known) objects in the universe is not a quasar but an ordinary galaxy, whose feeble light has been magnified significantly by the action of an intervening gravitational lens. We may henceforth need to rely upon these "intergalactic" telescopes to peer where (and when) ordinary telescopes cannot reach, and thus reveal the future holders of the cosmic distance record.

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